

Example: Microsatellite data set

POPULATION SIZE, MIGRATION, DIVERGENCE, ASSIGNMENT, HISTORY

Bayesian inference using the structured coalescent

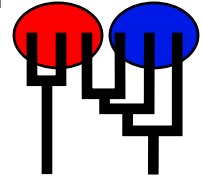
Migrate-n version 5.0.7 [May-01-2025]

Compiled for PARALLEL computer architectures

One master and 10 compute nodes are available.

Program started at Mon Jul 28 11:00:49 2025

Program finished at Mon Jul 28 11:01:09 2025 [Runtime:0000:00:00:20]



Options

Datatype: Microsatellite data [Brownian motion]
Missing data: not included

Inheritance multipliers in use for Thetas:
All loci use an inheritance multiplier of 1.0

Data set was subsampled: used a random sample of size:
5 and seed 13

Random number seed: (from parmfile) 310705631

Start parameters:

Theta values were generated Using a percent value of the prior

M values were generated Using a percent value of the prior

Connection matrix:

m = average (average over a group of Thetas or M,

s = symmetric migration M, S = symmetric 4Nm,

0 = zero, and not estimated,

* = migration free to vary, Thetas are on diagonal

d = row population split off column population, D = split and then migration

Population	1	2
1 population_num	*	0
2 population_num	D	*

1 population_numb growing/shrinking
 2 population_numb constant size

Order of parameters:

1	Θ_1	<displayed>
2	Θ_2	<displayed>
3	$M_{1 \rightarrow 2}$	<displayed>
4	$\Delta_{1 \rightarrow 2}$	<displayed>
5	$\sigma_{1 \rightarrow 2}$	<displayed>
6	g_1	<displayed>

Mutation rate among loci:

Mutation rate is constant for all loci

Analysis strategy:

Bayesian inference

-Population size estimation:

Exponential Distribution

-Population growth estimation: Growth [Exponential]

-Geneflow estimation:

Exponential Distribution

-Divergence time estimation:

Normal Distribution (mean and standard dev.)

Proposal distributions for parameter

Parameter	Proposal
Theta	Slice sampling
M	Slice sampling
Divergence	Metropolis sampling
Divergence Spread	Metropolis sampling
Growth	Metropolis sampling
Genealogy	Metropolis-Hastings

Prior distribution for parameter

Parameter	Prior	Minimum	Mean*	Maximum	Delta	Bins	UpdateFreq	
1 Theta	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
2 Theta	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
3 M	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
4 Splittime mean	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
5 Splittime mean	* *	Uniform	0.000000	10.00	20.00	2.000000	1500	0.08333
6 Growth	* *	Uniform	-100.000000	0.000	100.0	20.000000	1500	0.08333

[* * means priors were set globally]

Posterior distribution:

Parameter values were collected using MCMC, these values were then used to generate the posterior histograms using KERNEL SMOOTHING (window=41) and subsequent SAVITZKY-GOLAY SMOOTHING (window=41) for combination over loci

Markov chain settings:	Long chain
Number of chains	1
Recorded steps [a]	200
Increment (record every x step [b])	1
Number of concurrent chains (replicates) [c]	2
Visited (sampled) parameter values [a*b*c]	400
Number of discard trees per replicate (burn-in * b)	50

Multiple Markov chains:				
Static heating scheme		4 chains with	temperatures	
	1000000.00	3.00	1.50	1.00
		Swapping interval is 1		

Print options:	
Data file:	infile.msat
	parmfile.msat
Haplotyping is turned on:	NO
Output file:	outfile_msat
Posterior distribution raw histogram file:	bayesfile_msat
Raw data from the MCMC run:	bayesallfile_msat.gz
Print data:	No
Print genealogies [only some for some data type]:	None

Bayesian Analysis: Posterior distribution table

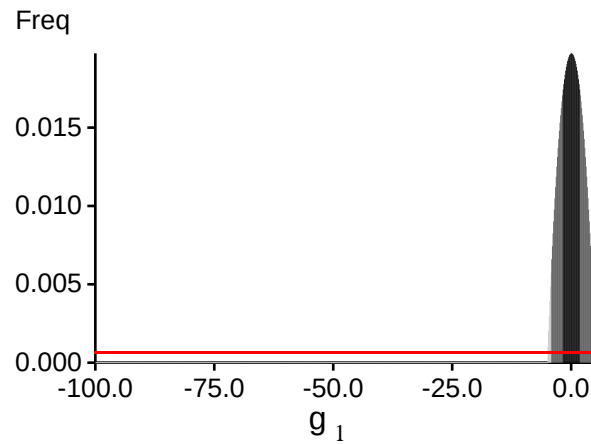
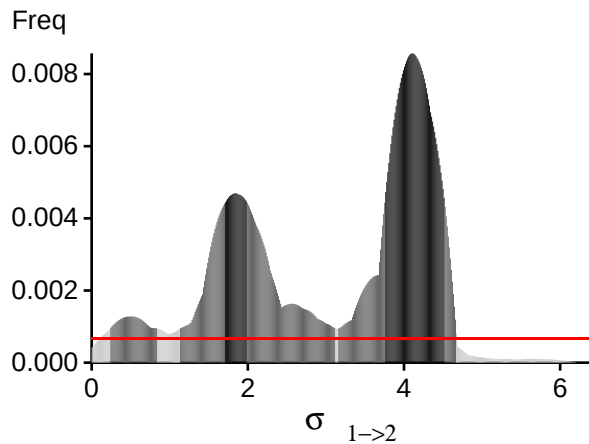
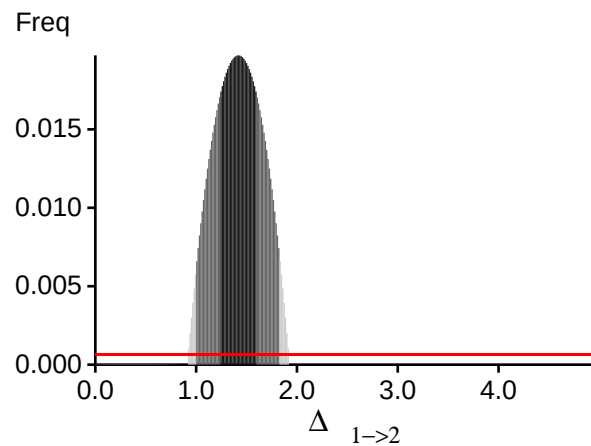
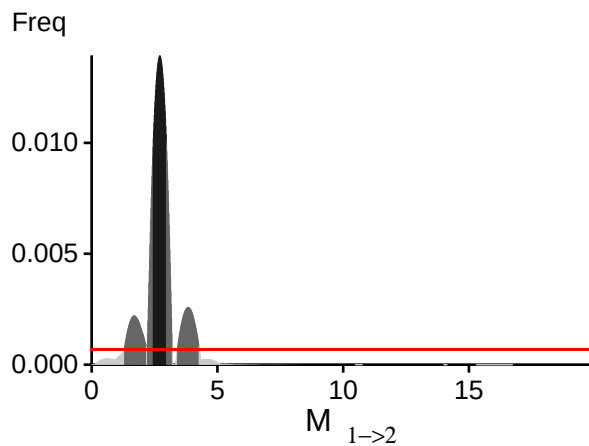
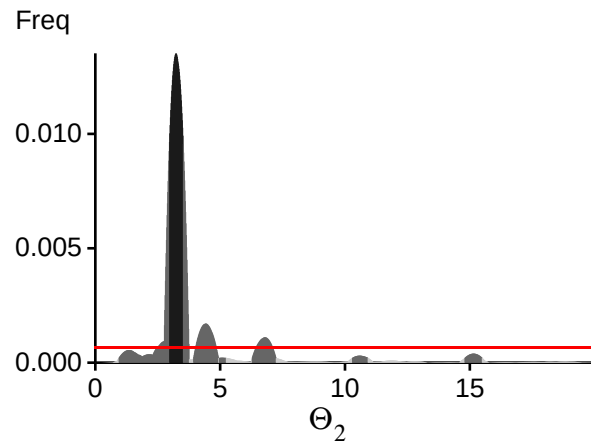
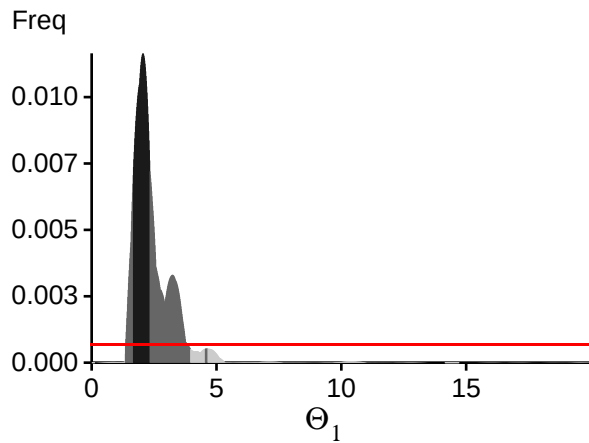
Locus	Parameter	2.5%	25.0%	Mode	75.0%	97.5%	Median	Mean
1	Θ_1	1.32000	1.65333	2.26000	2.98667	3.28000	2.98000	6.01681
1	Θ_2	4.68000	5.06667	5.44667	5.96000	6.44000	9.04667	10.00313
1	$M_{1 \rightarrow 2}$	1.013	2.133	2.820	3.307	3.707	2.687	2.755
1	$D_{1 \rightarrow 2}$	0.00000	0.18667	0.44667	0.81333	2.54667	1.15333	1.14010
1	$S_{1 \rightarrow 2}$	1.58667	3.12000	3.78000	4.17333	4.64000	3.50000	3.30249
1	g_1	-4.13333	-1.73333	0.20000	2.00000	4.40000	0.33333	0.23526
2	Θ_1	2.70667	2.86667	3.43333	4.06667	5.32000	6.62000	8.16202
2	Θ_2	3.38667	3.61333	4.08667	4.52000	4.82667	4.27333	5.81335
2	$M_{1 \rightarrow 2}$	1.080	2.853	3.273	3.813	5.467	5.607	7.406
2	$D_{1 \rightarrow 2}$	0.00000	1.52000	2.11333	2.41333	3.32000	2.38000	3.36526
2	$S_{1 \rightarrow 2}$	4.26667	4.42667	4.76667	5.18667	6.01333	3.38000	3.23966
2	g_1	-3.73333	-1.20000	0.73333	2.53333	4.93333	0.86667	0.69790
3	Θ_1	0.13333	0.21333	0.62000	1.02667	2.45333	7.80667	8.94243
3	Θ_2	0.89333	1.05333	1.50000	1.93333	5.18667	4.12667	6.89097
3	$M_{1 \rightarrow 2}$	0.120	0.427	0.980	1.280	2.813	1.713	2.407
3	$D_{1 \rightarrow 2}$	2.05333	2.17333	2.48667	2.92000	4.68000	3.12667	3.21469
3	$S_{1 \rightarrow 2}$	0.92000	0.97333	1.40667	1.80000	1.96000	2.99333	3.06076
3	g_1	-6.80000	-2.26667	-0.06667	2.26667	5.73333	0.06667	-0.25645
4	Θ_1	2.58667	2.81333	3.27333	3.58667	6.38667	4.24667	4.38434
4	Θ_2	0.90667	1.06667	1.46000	1.81333	2.02667	5.87333	7.14114
4	$M_{1 \rightarrow 2}$	0.000	0.120	0.553	1.080	1.667	2.807	2.842
4	$D_{1 \rightarrow 2}$	1.17333	1.46667	1.79333	2.08000	3.01333	2.04667	2.07182
4	$S_{1 \rightarrow 2}$	0.40000	0.40000	0.87333	1.96000	1.96000	1.94000	2.53022
4	g_1	-4.13333	-1.73333	0.20000	2.00000	4.40000	0.33333	0.13600
5	Θ_1	1.29333	2.10667	2.83333	3.34667	4.50667	2.87333	2.88776
5	Θ_2	9.60000	9.74667	10.91333	11.65333	12.06667	10.92667	10.66690
5	$M_{1 \rightarrow 2}$	0.600	3.427	4.007	4.427	6.280	3.900	4.573
5	$D_{1 \rightarrow 2}$	0.00000	0.81333	1.27333	1.62667	2.01333	1.15333	1.11173
5	$S_{1 \rightarrow 2}$	0.02667	0.34667	0.75333	1.52000	1.81333	1.40667	1.54509
5	g_1	-4.26667	-1.33333	0.60000	2.53333	4.93333	0.73333	0.57672
6	Θ_1	1.16000	2.41333	3.51333	3.88000	5.12000	3.48667	5.22930

Locus	Parameter	2.5%	25.0%	Mode	75.0%	97.5%	Median	Mean
6	Θ_2	2.69333	2.82667	3.24667	3.69333	3.77333	7.63333	9.47997
6	$M_{1 \rightarrow 2}$	0.667	0.933	1.327	1.653	4.587	3.020	3.367
6	$D_{1 \rightarrow 2}$	2.02667	2.24000	2.68667	3.12000	3.25333	2.12667	1.88546
6	$S_{1 \rightarrow 2}$	3.06667	3.38667	3.76667	4.13333	4.68000	3.46000	3.17720
6	g_1	-4.13333	-1.86667	0.06667	1.86667	4.26667	0.20000	0.08589
7	Θ_1	0.17333	0.42667	0.90000	1.36000	2.78667	1.98000	2.87531
7	Θ_2	0.90667	2.62667	3.48667	4.88000	5.49333	4.00667	4.62757
7	$M_{1 \rightarrow 2}$	0.427	1.320	1.767	2.293	4.400	3.300	3.526
7	$D_{1 \rightarrow 2}$	0.64000	1.05333	1.35333	1.84000	2.70667	2.19333	2.74778
7	$S_{1 \rightarrow 2}$	2.17333	2.48000	2.84667	3.18667	3.38667	1.84667	1.75818
7	g_1	-4.53333	-2.00000	-0.06667	1.73333	4.26667	0.06667	-0.13131
8	Θ_1	1.34667	1.74667	2.47333	3.14667	5.40000	4.38000	6.32107
8	Θ_2	0.45333	2.20000	2.74000	3.28000	3.94667	3.10000	4.31385
8	$M_{1 \rightarrow 2}$	1.013	1.453	1.820	2.160	4.467	2.887	2.999
8	$D_{1 \rightarrow 2}$	1.41333	1.88000	2.12667	2.36000	2.56000	1.34000	1.25393
8	$S_{1 \rightarrow 2}$	0.40000	1.72000	1.95333	2.57333	3.28000	2.07333	1.99830
8	g_1	-4.13333	-1.73333	0.20000	2.00000	4.26667	0.33333	0.14433
9	Θ_1	1.06667	2.74667	3.34000	3.72000	3.97333	3.22000	4.50509
9	Θ_2	10.80000	11.16000	11.32667	11.49333	13.24000	12.54000	12.68056
9	$M_{1 \rightarrow 2}$	0.160	0.240	0.687	1.147	2.147	1.527	2.032
9	$D_{1 \rightarrow 2}$	2.64000	3.00000	3.56667	4.01333	4.25333	3.70000	3.77994
9	$S_{1 \rightarrow 2}$	0.21333	0.86667	1.39333	1.72000	2.36000	2.03333	2.41434
9	g_1	-4.26667	-1.86667	0.06667	1.86667	4.26667	0.20000	0.00020
10	Θ_1	1.70667	2.13333	2.64667	3.33333	5.42667	3.15333	4.41417
10	Θ_2	3.60000	4.86667	5.28667	5.61333	6.00000	8.58000	9.41877
10	$M_{1 \rightarrow 2}$	5.053	5.187	5.527	5.867	6.000	5.113	5.088
10	$D_{1 \rightarrow 2}$	0.65333	2.37333	2.66000	3.05333	3.73333	2.28667	2.18961
10	$S_{1 \rightarrow 2}$	0.00000	0.14667	0.43333	0.72000	1.29333	2.75333	2.04220
10	g_1	-4.26667	-1.73333	0.20000	2.00000	4.53333	0.33333	0.20781
All	Θ_1	1.33333	1.65333	2.06000	2.33333	3.96000	2.20667	2.49637
All	Θ_2	0.93333	2.96000	3.23333	3.52000	3.76000	3.34000	4.15767
All	$M_{1 \rightarrow 2}$	2.200	2.427	2.713	2.973	3.200	2.727	2.736
All	$D_{1 \rightarrow 2}$	0.98667	1.22667	1.42000	1.60000	1.82667	1.43333	1.41909
All	$S_{1 \rightarrow 2}$	3.14667	3.74667	4.11333	4.52000	4.66667	3.55333	3.01069
All	g_1	-4.26667	-1.86667	0.06667	1.86667	4.26667	0.20000	0.06667

Citation suggestions:

- Beerli P., 2006. Comparison of Bayesian and maximum-likelihood inference of population genetic parameters. *Bioinformatics* 22:341-345
- Beerli, P., H. Ashki, S. Mashayekhi, and M. Palczewski, 2022. Population divergence time estimation using individual lineage label switching. *G3 Genes & Genomes & Genetics*, 12(4), 02 2022.
- Beerli P., 2009. How to use MIGRATE or why are Markov chain Monte Carlo programs difficult to use? In *Population Genetics for Animal Conservation*, G. Bertorelle, M. W. Bruford, H. C. Hauffe, A. Rizzoli, and C. Vernesi, eds., vol. 17 of *Conservation Biology*, Cambridge University Press, Cambridge UK, pp. 42-79.
- Beerli, P., S. Mashayekhi, M. Sadeghi, M. Khodaei, and K. Shaw, 2019. Population genetic inference with migrate. *Current Protocols in Bioinformatics*, 68(1):e87.

Bayesian Analysis: Posterior distribution over all loci



Log-Probability of the data given the model (marginal likelihood)

Use this value for Bayes factor calculations:

$BF = \text{Exp}[\ln(\text{Prob}(D \mid \text{thisModel})) - \ln(\text{Prob}(D \mid \text{otherModel}))]$

or as $LBF = 2 (\ln(\text{Prob}(D \mid \text{thisModel})) - \ln(\text{Prob}(D \mid \text{otherModel})))$

shows the support for thisModel]

Locus	TI(1a)	BTI(1b)	HS(3)
1	-679.16	-253.61	-515.65
2	-31230.74	-5040.64	-127.02
3	-708.21	-154.68	-111.19
4	-5534.51	-1124.14	-2074.54
5	-32.17	-25.01	-45.25
6	-1014.98	-266.50	-182.59
7	-81.57	-42.89	-83.57
8	-98.82	-70.74	-82.77
9	-17200.15	-2825.17	-81.50
10	-560.78	-146.62	-165.69
All	-57071.54	-9880.46	-3400.22

(1a) TI: Thermodynamic integration: $\log(\text{Prob}(D \mid \text{Model}))$: Good approximation with many temperatures

(1b) BTI: Bezier-approximated Thermodynamic integration: when using few temperatures USE THIS!

(3) HS: Harmonic mean approximation: Overestimates the marginal likelihood, poor variance

[Scaling factor = 69.543961]

Citation suggestions:

Beerli P. and M. Palczewski, 2010. Unified framework to evaluate panmixia and migration direction among multiple sampling locations, *Genetics*, 185: 313-326.

Palczewski M. and P. Beerli, 2014. Population model comparison using multi-locus datasets.

In M.-H. Chen, L. Kuo, and P. O. Lewis, editors, *Bayesian Phylogenetics: Methods, Algorithms, and Applications*, pages 187-200. CRC Press, 2014.

Acceptance ratios for all parameters and the genealogies

Parameter	Accepted changes	Ratio
Θ_1	342/342	1.00000
Θ_2	340/340	1.00000
$M_{1 \rightarrow 2}$	337/337	1.00000
$\Delta_{1 \rightarrow 2}$	261/283	0.92226
$\sigma_{1 \rightarrow 2}$	261/283	0.92226
$\Delta_{1 \rightarrow 1}$	288/307	0.93811
$\sigma_{1 \rightarrow 1}$	288/307	0.93811
g_7	25/333	0.07508
Genealogies	1003/333	3.01201

MCMC-Autocorrelation and Effective MCMC Sample Size

Parameter	Autocorrelation	Effective Sampe Size
Θ_1	0.94663	110.53
Θ_2	0.90905	191.15
$M_{1 \rightarrow 2}$	0.89883	217.14
$\Delta_{1 \rightarrow 2}$	0.96191	77.73
$\sigma_{1 \rightarrow 2}$	0.95002	102.90
g_1	0.96685	71.58
Genealogies	0.00000	0.00

Static temperatures during the run using standard heating

Chain	Temperatures	log(marginal likelihood)	log(mL_steppingstone)
1	1.00000	-58.41500	-62.79522
2	1.00000	-77.82783	-58.34561
3	1.00000	-117.25614	-41.29474
4	1.00000	-33836.16936	-14.86119

Potential Problems

This section reports potential problems with your run, but such reporting is often not very accurate. With many parameters in a multilocus analysis, it is common that some parameters for some loci will not be informative. These parameters then trigger suggestions to increase the prior range that are not sensible. Do not blindly follow the suggestions given. If some parameters are flagged, inspect the tables carefully and judge whether an action is required. Suppose you run a Bayesian inference with sequence data for macroscopic species. In that case, there is rarely the need to increase the prior for Theta beyond 0.1. If you use microsatellites data, it is rather common that your prior distribution for Theta should have a range from 0.0 to 100 or more. With many populations (>3), it is also very common that some migration routes are estimated poorly because the data contains little or no information for that route. Increasing the prior range will not help in such situations, but reducing the number of parameters may help.

No warning was recorded during the run